

Hossam Abu-elez Fathy

Machine Learning Engineer | Data Analyst | Data Scientist
Bachelor of Computer Science & Artificial Intelligence

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🐙 GitHub

🌐 LinkedIn

🌐 Portfolio

PROFILE

Computer Science and Artificial Intelligence student with hands-on experience in Machine Learning, Data Analysis, and Data Science. Experienced in building end-to-end AI solutions using Python, Scikit-Learn, SQL, and data visualization tools. Developed machine learning models achieving up to 99.46% accuracy on real-world datasets. Seeking internships and entry-level opportunities in Machine Learning, Data Analytics, and Data Science.

EDUCATION

•Artificial Intelligence Sciences

New Mansoura University

Oct 2023 – Jun 2027

CGPA: 3.4 / 4.0

EXPERIENCE

•Machine Learning Engineering Intern

FlyRank AI

Jul 2026 – Present

Remote

- Developing machine learning solutions for real-world AI applications.
- Building, evaluating, and improving ML models using Python.
- Collaborating with engineering teams on AI-driven projects.

•Data Analytics & BI Trainee

Information Technology Institute (ITI)

Jul 2026 – Present

Mansoura

- Building dashboards using Power BI and Tableau.
- Writing SQL queries and performing data analysis using Excel.
- Learning Data Warehousing and Business Intelligence concepts.

•Machine Learning Trainee

Creativa Innovation Hub

Jun 2025 – Aug 2025

Mansoura

- Preprocessed datasets and built ML models using Scikit-Learn.
- Applied feature engineering, classification, and regression techniques.
- Visualized data insights using Matplotlib and Plotly.

TECHNICAL SKILLS

Programming: Python, C++

Machine Learning: Supervised Learning, Unsupervised Learning, Deep Learning, Neural Networks, Machine Learning Algorithms, Model Evaluation, Model Selection, Feature Engineering, Data Preprocessing

Libraries: Scikit-Learn, Pandas, NumPy, Matplotlib, Plotly

Data Analysis: Excel, Power BI, EDA, Data Cleaning, Data Visualization

Databases: SQL, MySQL, PostgreSQL

Software Engineering: REST APIs, Git, GitHub, Object-Oriented Programming, Software Engineering Methodologies, Full-Stack Development

Tools: Git, GitHub, Markdown, Jupyter Notebook, VS Code, Kaggle

Soft Skills: Problem Solving, Communication, Team Collaboration, Analytical Thinking, Presentation, Adaptability, Continuous Learning

PROJECTS

•Analytics of Jumia Product Reviews

2026

Data Analytics & Business Intelligence Project

- Analyzed **3,809** Jumia product reviews covering 877 unique products using SQL Server, Python, and Power BI.
- Performed data exploration, validation, cleaning, and text preprocessing using SQL Server, Pandas, and NumPy, including Arabic and English review analysis.
- Built an interactive Power BI dashboard with data modeling, KPIs, filters, drill-through, product performance, and customer feedback analysis.
- **Technologies:** Python, Pandas, NumPy, SQL Server, Power BI, Excel, Text Analytics.

•MedOptima AI Platform

2025

AI Healthcare Decision Support System

- Developed an end-to-end AI healthcare decision support platform using Neural Networks, Knowledge-Based Systems, and Optimization Algorithms.
- Achieved **99.46%** test accuracy on a Kaggle dataset containing 41 diseases and 117 symptoms.
- Performed data preprocessing, feature engineering, hyperparameter tuning, and model evaluation.
- **Technologies:** Python, Scikit-Learn, SQL, REST APIs, Neural Networks, Full-Stack Development.

•AI Optimization Research Project - FTT Algorithm

2025

Research and Optimization

- Designed and implemented the Football Team Training (FTT) Optimization Algorithm.
- Solved supply chain optimization problems and compared the proposed algorithm with existing methods.
- Improved convergence speed and solution quality.

•Air Quality Prediction using Machine Learning

2024 – 2025

Machine Learning Regression Project

- Collected, cleaned, and preprocessed environmental datasets for predictive modeling.
- Built and evaluated regression models using Python and Scikit-Learn.
- Performed EDA and created interactive visualizations using Plotly and Matplotlib.

•ALU 4-bit Digital Design Project

2024

Computer Architecture Project

- Designed and implemented a 4-bit ALU supporting arithmetic and logical operations.
- Simulated and tested the ALU using digital logic design tools.

•Computer Architecture & Assembly - 8085 & 8086

2024

Assembly Programming

- Implemented and analyzed assembly programs using 8085 and 8086 instruction sets.
- Analyzed processor architecture, memory organization, and instruction execution.

•Airline Reservation System

2023 – 2024

Full Software Engineering Project

- Designed system requirements, UML diagrams, and system architecture.
- Applied Object-Oriented Programming (OOP) principles and software engineering methodologies.
- Integrated a relational database for efficient data management.

CERTIFICATES

•Freelancing Basics

Sep 2026

•Find Your Freelance Fit

Sep 2026

•Getting Started as a Freelancer (Upwork)

Sep 2026

•Getting Started as a Freelancer (Khamsat)

Sep 2026

•Getting Started as a Freelancer (Mostaql)

Sep 2026

•Machine Learning Training

Creativa Innovation Hub
Jun 2025

•Transact SQL Queries Using SQL Server

Mahara Tech
Mar 2025

•Communicate Effectively on GitHub using Markdown

Microsoft Learn
Apr 2024

LEADERSHIP & VOLUNTEERING

- **AI & Programming Student Mentor** Supporting Students and Beginners *Present*
 - Mentored students in AI, Python, and programming fundamentals.
 - Guided beginners in problem-solving and technical skill development.
- **Tech Event Organizer** AI & Technology Workshops *Present*
 - Organized AI and technology workshops for students.
 - Coordinated event logistics and supported successful workshop execution.

LANGUAGES

- **Arabic:** Native
- **English:** Proficient